

Risultati per paper relativi alla fisica su inspire Hep

QUERY Utilizzata su INSPIRE HEP:

<https://inspirehep.net/api/literature?q=t+quantum+mechanics+and+date+2016-%3E2026&sort=mostcited&size=10>

!!! Recupera 10 paper sulla meccanica quantistica dal database di Inspire Hep, data paper compresa tra 2016 e 2026. In ordine decrescente di numero di citazioni.

Siccome alcuni paper hanno tante figure e ogni figura significa una chiamata alle Api, ho testato solamente i paper con un numero contenuto di figure.

TITOLO PAPER ANALIZZATO: **Sachdev–Ye–Kitaev model as Liouville quantum mechanics**

INSPIRE HEP OUTPUT:

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----- ANALISI DEL PAPER SENZA PIPELINE VISIVA -----

VERSIONE PDF:

comando lanciato: python main.py analyze --pdf downloaded_pdfs/"1473712_Sachdev–Ye–Kitaev model as Liouville quantum mecha.pdf" --model gemini

----- Gemini -----

Model used: gemini-3.6-flash

Output JSON:

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{
  "problem_addressed": "The paper addresses the long-time and low-frequency behavior of correlation functions in the Sachdev-Ye-Kitaev (SYK) model of N randomly interacting Majorana fermions, where soft time-reparameterization fluctuations break conformal invariance.",
  "main_contributions": [
    "Mapped the soft reparameterization Goldstone modes of the SYK model onto Liouville quantum mechanics using a non-linear variable transformation.",
    "Proved that all zero-temperature connected correlation functions exhibit a universal power-law decay of  $\sim t^{-3/2}$  at long times  $t \gg N \ln N$ .",
    "Showed that the single-particle Green function exhibits a zero-bias anomaly at low frequencies, scaling as  $|\omega|^{1/2}$  instead of the mean-field  $|\omega|^{-1/2}$  scaling.",
    "Determined the effective self-consistent single-particle level spacing scale as  $\sim J / (N \ln N)$ .",
    "Supported theoretical findings with exact diagonalization calculations for N = 24 Majorana fermions."
  ],
  "methodology": "The authors used a path-integral formulation with replicas and stationary-phase approximations to isolate soft reparameterization Goldstone modes  $f(\tau)$ . By applying the non-linear substitution  $f(\tau) = e^{i\phi(\tau)}$ , the soft-mode Schwarzian effective action was reduced to a quadratic form in  $\phi(\tau)$ . Correlation functions were then evaluated as quantum-mechanical matrix elements using the spectral decomposition of Schrödinger wavefunctions in a

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Liouville exponential potential (e^{λ}). Exact diagonalization on $N = 24$ Majorana fermions was performed to compute the Matsubara Green function numerically.",

"datasets_used": [],

"experiments": "Numerical exact diagonalization of the Hamiltonian for $N = 24$ Majorana fermions with fixed Gaussian random couplings to calculate the low-energy Matsubara Green function $G(\omega)$ and compare with the analytical spectral representation.",

"results": "At long times $t \gg N \ln N$, all correlation functions cross over from standard mean-field power-law behavior to a universal $t^{-3/2}$ decay. For the single-particle Green function, the density of states transitions from $\omega^{1/2}$ at higher energies to an $\omega^{1/2}$ suppression at low frequencies. Exact diagonalization results for $N = 24$ confirm the vanishing of $G(\omega)$ at small energies and qualitative agreement with the analytical crossover.",

"limitations": "The finite-size numerical simulation ($N = 24$) cannot fully resolve the exact $\omega^{1/2}$ exponent due to the narrow energy range between the many-body level spacing ($\sim e^{-(N/2)}$) and the single-body scale ($\sim 1/(N \ln N)$). Additionally, the local approximation to the action holds specifically for 4-body interactions ($q = 4$).",

"future_work": "Exploring the concrete manifestations of this infrared universal behavior in the holographic $\text{NAdS}_2/\text{NCFT}_1$ correspondence.",

"relevant_elements": "Equation 1: SYK Hamiltonian; Equation 13: Local soft-mode Schwarzian action $S[f]$; Equation 14: Non-linear transformation $f'(\omega) = e^{\lambda(\omega)}$; Equation 19: Liouville Schrödinger equation; Equation 24: Spectral representation of Green function $G(\omega)$; Figure 1: Numerical vs analytic Matsubara Green function $G(\omega)$ for $N = 24$.",

"visual_analysis_findings": []

}

VERSIONE LATEX:

comando lanciato: python main.py analyze --latex downloaded_latex/"1473712_Sachdev–Ye–Kitaev model as Liouville quantum mecha_source.tar.gz"

Model used: nvidia/nemotron-3-nano-omni-30b-a3b-reasoning:free

Output JSON:

{

"problem_addressed": "The paper shows that including soft reparameterization modes in the SYK model maps its long-time behavior to Liouville quantum mechanics, yielding universal $t^{-3/2}$ decay of correlation functions.",

"main_contributions": [

"Demonstrate mapping of SYK model with soft reparameterization modes to Liouville quantum mechanics",

"Derive universal long-time $t^{-3/2}$ scaling for all correlation functions and zero-bias anomaly $\omega^{1/2}$ scaling",

"Provide exact diagonalization results for $N=24$ confirming analytical predictions",

"Determine effective single-particle level spacing $\Delta \sim J/(N \log N)$ "

],

"methodology": "They use replica averaging, derive the effective action for reparameterization modes, perform exact functional integration over ϕ , map to Liouville quantum mechanics, and compute Green functions via spectral representation, verifying $t^{-3/2}$ scaling.",

"datasets_used": [

"Exact diagonalization of SYK model for $N=24$ Majorana fermions"

```

],
"experiments": "Exact diagonalization of the SYK model for  $N=24$  to compute Matsubara Green functions and compare with analytical predictions.",
"results": "Long-time correlation functions decay as  $\omega^{-3/2}$  for  $\omega \ll N \ln N$ ; single-particle Green function exhibits zero-bias anomaly with  $\omega^{1/2}$  scaling at low energy; exact diagonalization confirms  $\omega^{-3/2}$  scaling and  $\omega^{1/2}$  behavior; effective level spacing  $\omega \sim J/(N \ln N)$ ; four-point function shows  $M^{1/2} \omega^{-3/2}$  decay for  $\omega \ll M$ ; universal long-time decay  $\omega^{-3/2}$  for arbitrary operators.",
"limitations": "Finite many-body level spacing limits resolution of small-energy scaling; analysis restricted to  $J > N \ln N$  and does not address holographic duality.",
"future_work": "Explore holographic correspondence between SYK and Liouville quantum mechanics; refine treatment of massive modes and soft mode divergences.",
"relevant_elements": [
  "Equation (\ref{eq:model}): SYK Hamiltonian",
  "Equation (\ref{eq:distribution}): Distribution of coupling constants",
  "Equation (\ref{eq:action}): Effective action",
  "Equation (\ref{eq:Dyson}): Dyson equations",
  "Equation (\ref{eq:bare-solution}): Bare Green function",
  "Equation (\ref{eq:soft-manifold}): Reparameterization manifold",
  "Equation (\ref{eq:soft-action}): Schwarzian action",
  "Equation (\ref{eq:phi-action}):  $\phi$ -action",
  "Equation (\ref{eq:two-point}): Two-point Green function",
  "Equation (\ref{eq:quantum-mechanics}): Quantum mechanical representation",
  "Equation (\ref{eq:Schrödinger}): Schrödinger equation",
  "Equation (\ref{eq:eigenfunction}): Eigenfunction",
  "Equation (\ref{eq:matrix-elements}): Matrix elements",
  "Equation (\ref{eq:G-semi-final}): Final Green function expression",
  "Equation (\ref{eq:M_2}): Spectral density",
  "Equation (\ref{eq:four-point-action}): Four-point action",
  "Equation (\ref{eq:polarization-semi-final}): Polarization operator",
  "Equation (\ref{eq:2p}): 2p-point function",
  "Equation (\ref{eq:universal}): Universal long-time decay",
  "Equation (\ref{eq:self-consistent}): Self-consistent equation for  $M$ ",
  "Equation (\ref{eq:Pi_log}): Logarithmic polarization operator",
  "Equation (\ref{eq:S2_reg}): Regularized soft-mode action",
  "Exact diagonalization algorithm for  $N=24$  SYK model",
  "Figure 1: Numerical versus analytic Green function comparison"
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"visual_analysis_findings": []
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-----PIPELINE GRAFICA-----

Versione PDF:

Comando: python main.py full-analyze --pdf downloaded_pdfs/"1473712_Sachdev–Ye–Kitaev model as Liouville quantum mecha.pdf" --model gemini

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--- Figure 1/2 ---

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[+] Success: detected 'continuous' plot with 2 distinct curve(s).

[~] Starting PySR (Symbolic Regression) with 20 iterations...

[+] Mathematical formula discovered with success!

-> Expression: $x_0 - (-0.56305695) \cdot x_0 / ((x_0 - 1 \cdot (-0.05222697)) \cdot (x_0 - 1 \cdot (-0.05222697)))$

-> Complexity: 13 | R^2 (fit): 0.9924 | R^2 (holdout): 0.9726

[~] Starting PySR (Symbolic Regression) with 20 iterations...

[+] Mathematical formula discovered with success!

-> Expression: $(0.9497001 - \exp(x_0 \cdot (-45.737568))) / (-x_0 \cdot x_0 + x_0 + \exp(x_0 - 1 \cdot 1.331508))$

-> Complexity: 17 | R^2 (fit): 0.9681 | R^2 (holdout): 0.9763

--- Figure 2/2 ---

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[+] Enriched paper content with analysis of 1/2 figure(s).

Output JSON:

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{
  "problem_addressed": "The paper addresses the long-time behavior of the Sachdev-Ye-Kitaev (SYK) model by showing that proper inclusion of soft reparameterization modes reduces the model to Liouville quantum mechanics, leading to universal power-law decay of correlation functions with exponent  $-3/2$  at times  $\sim N \ln N$ , and a zero-bias anomaly in the single-particle Green function scaling as  $\omega^{1/2}$  at low energies.",
  "main_contributions": [
    "Mapping the effective action of reparameterization Goldstone modes in the SYK model to Liouville quantum mechanics via a nonlinear variable transform.",
    "Demonstrating universal long-time decay  $\sim t^{-3/2}$  for arbitrary operator correlation functions at zero temperature.",
    "Identifying the zero-bias anomaly: single-particle density of states crosses over from  $\omega^{-1/2}$  to  $\omega^{1/2}$  suppression at low energies.",
```

"Providing exact diagonalization results for $N=24$ Majorana fermions supporting the analytical predictions.",

"Self-consistently determining the effective single-particle mean-level spacing $J/(N \log N)$.",

],

"methodology": "The paper uses a path integral formulation with replica trick and disorder averaging to derive a bilocal action. Saddle point equations yield a conformal solution that spontaneously breaks reparameterization symmetry down to $SL(2, \mathbb{R})$. Soft Goldstone modes are described by an effective non-local action, which is approximated by a local Schwarzian derivative action. A nonlinear change of variables $f(\tau) = e^{\phi(\tau)}$ maps the soft-mode action to a free particle action with a Liouville potential. Correlation functions are computed as expectation values in Liouville quantum mechanics via spectral decomposition using modified Bessel function eigenstates. Exact diagonalization of the SYK Hamiltonian for $N=24$ is used for numerical verification.",

"datasets_used": [

"Exact diagonalization data for the SYK model with $N=24$ Majorana fermions (single disorder realization).",

],

"experiments": "The analytical predictions for the two-point Green function $G(\omega)$ and its scaling behavior are compared with exact diagonalization results for $N=24$. The Matsubara Green function is evaluated via Lehmann representation. The four-point function (polarization operator) is analyzed for different time orderings, and the self-consistent equation for the mass parameter M is solved. The universal long-time exponent is verified for general $2p$ -point functions.",

"results": [

"Zero-temperature correlation functions decay universally as $\omega^{-3/2}$ for $\omega \ll N \ln N$.",

"Single-particle Green function: $G(\omega) \sim |\omega|^{-1/2}$ for $\omega \ll M$; $G(\omega) \sim M/|\omega|^{3/2}$ for $\omega \gg M$. In energy: $G(\omega) \sim |\omega|^{-1/2}$ for $\omega \ll M^{-1}$; $G(\omega) \sim M |\omega|$ for $\omega \gg M^{-1}$.",

"Polarization operator $G_{\{\omega, 0\}} G_{\{0, \omega\}} \sim |\omega|^{-1}$ for $\omega \ll M$; $\sim M^{1/2} |\omega|^{-3/2}$ for $\omega \gg M$.",

"General $2p$ -point functions: $G^p_{\{\omega, 0\}} \sim M^{(3-p)/2} |\omega|^{-3/2}$ at long times.",

"Self-consistent mass parameter: $M = (b^2/(32J)) N \log(JM)$ $(b^2/(32J)) N \log N$, with $b = (4\pi)^{-1/4}$.",

"Effective single-particle level spacing $J/(N \log N)$.",

"Exact diagonalization for $N=24$ shows qualitative agreement; finite-size effects prevent clear resolution of the $\omega^{1/2}$ exponent at the smallest energies.",

],

"limitations": "The theory does not capture physics at energies below the many-body level spacing ($\sim e^{-N/2}$). For $N=24$, the energy window between the many-body level spacing and the one-body scale $M^{-1} \sim 1/(N \log N)$ is too narrow to clearly resolve the predicted $\omega^{1/2}$ scaling. The analysis ignores replica off-diagonal fluctuations, justified by self-averaging at energies above the many-body level spacing. The local approximation to the soft-mode action is valid only for $q=4$ interaction; for generic q it remains non-local.",

"future_work": "Exploring the consequences of the universal long-time behavior within the holographic interpretation of the SYK model ($NAdS_2/NCFT_1$ correspondence). Extending the analysis to models with generic q -body interactions where the soft-mode action is inherently non-local.",

"relevant_elements": [

"Equation (1): SYK Hamiltonian with Gaussian random couplings.",

"Equation (2): Distribution of coupling constants $J_{\{ijkl\}}$.",
 "Equation (6): Bilocal action after disorder averaging and Hubbard-Stratonovich transformation.",
 "Equation (7): Self-consistent Dyson equations for G and Σ .",
 "Equation (8): Conformal saddle-point solution for G and Σ in time domain.",
 "Equation (10)-(11): Reparameterized Green function and self-energy under $f(\omega) \rightarrow f(\omega - i0^+)$.",
 "Equation (12): Non-local soft-mode action $S[f]$.",
 "Equation (13): Local soft-mode action in Schwarzian derivative form with mass M .",
 "Equation (14): Variable transform $f'(\omega) = e^{\phi(\omega)}$.",
 "Equation (15): Quadratic action $S[\phi] = (M/2) \int d\omega (\phi')^2$.",
 "Equation (16): Two-point Green function as path integral with Liouville potential.",
 "Equation (18): Spectral decomposition of $G(\omega)$ in Liouville quantum mechanics.",
 "Equation (19): Liouville Schrödinger equation.",
 "Equation (20): Eigenfunctions $\psi_k(\omega) = N_k K_{\{2ik\}}(2M\omega) e^{\phi(\omega/2)}$.",
 "Equation (22): Final two-point Green function with spectral density $M^2(k)$.",
 "Equation (23): Spectral density $M^2(k) = (k \sinh(2\omega_0))/(2\omega_0^2) |1/4 + ik|^4$.",
 "Equation (24): Energy representation of Green function.",
 "Equation (25): Definition of four-point function.",
 "Equation (31): Polarization operator result with $M_4(k) = k \tanh(\omega_0 k)$.",
 "Equation (32): General 2p-point correlation function formula.",
 "Equation (33): General spectral function $M_{\{2p\}}(k)$.",
 "Equation (34): Universal long-time decay $\omega^{p-3/2} M^{\{(3-p)/2\}} |\omega|^{-3/2}$.",
 "Equation (35): Self-consistent equation for M .",
 "Figure 1: Numerical (exact diagonalization) vs analytical Matsubara Green function for $N=24$."],
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 "AI curve-fit approximation of Figure 1 (left panel), solid red line: $y = (0.9497001 - \exp(x_0 * (-45.737568))) / (-x_0 * x_0 + x_0 + \exp(x_0 - 1 * 1.331508))$, $R^2=0.968$ (not derived or stated by the paper's authors)."
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}

VERSIONE LATEX:

comando: python main.py full-analyze --latex downloaded_latex/"1473712_Sachdev–Ye–Kitaev model as Liouville quantum mecha_source.tar.gz" --model gemini

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[2/3] Extracting figures...

[~] Extracting images from LaTeX archive...

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--- Figure 1/2 ---

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[~] Computer vision analysis in progress: G2_plot.png

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--- Figure 2/2 ---

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[!] No usable figure analysis produced; proceeding with text-only content.

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 "problem_addressed": "Understanding the long-time behavior of correlation functions in the Sachdev-Ye-Kitaev (SYK) model and demonstrating their universal scaling properties arising from soft reparameterization modes",
 "main_contributions": [
 "Mapping the SYK model to Liouville quantum mechanics via soft reparameterization modes",
 "Deriving the soft mode action and showing it reduces to a Schwarzian derivative form",
 "Proving universal long-time decay of correlation functions as $\tau^{-3/2}$ ",
 "Providing exact diagonalization support for the theoretical predictions"
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 "methodology": "The paper introduces soft reparameterization modes (Goldstone modes) of the SYK model, maps them to Liouville quantum mechanics using a nonlinear transformation ϕ , and computes correlation functions by averaging over soft mode fluctuations. The approach involves identifying the effective action for the reparameterization modes and evaluating the resulting path integral.",
 "datasets_used": [],
 "experiments": "Numerical calculations of two-point Green functions (Figure G) comparing numerical and analytic results for $N=24$ Majorana fermions, exact diagonalization studies supporting the theoretical conclusions, and analytical evaluation of four-point functions.",
 "results": "Two-point Green functions decay as $|\tau-\tau'|^{-3/2}$ at long times. Universal scaling law $\langle G_{\tau,0^p} \rangle \propto M^{(3-p)/2}/|\tau|^{3/2}$ is established. The effective single-particle mean-level spacing is $\Delta \sim J/(N \log N)$. The soft mode action is $S[f] = M/2 \int d\tau (f''(\tau)/f'(\tau))^2$ with $M = b^2/(32J) \cdot N \log(JM)$.",
 "limitations": "The analysis assumes asymptotic slow time variations compared to the interaction strength J . The behavior at very small energies is influenced by the finite many-body level spacing ($\sim e^{-(N/2)}$), making precise agreement with $\epsilon^{1/2}$ scaling difficult for small N .",
 "future_work": "Exploring the holographic correspondence implications and extending the analysis to other regimes of the SYK model.",
 "relevant_elements": [
 "Equation (eq:soft-action): $S[f] = M/2 \int d\tau (f''(\tau)/f'(\tau))^2$ ",
 "Equation (eq:universal): $\langle G_{\tau,0^p} \rangle \propto M^{(3-p)/2}/|\tau|^{3/2}$ ",
 "Self-consistent equation: $M = b^2/(32J) \cdot N \log(JM)$ ",
 "Mapping to Liouville quantum mechanics via the transformation $f(\tau)$ "
],
 "visual_analysis_findings": []
}
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}